

LAB PROGRAMS (36-40)
COMPUTER VISION-ITA-0505
SLOT-D

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QUESTION-36

Write a Python function to subtract the background of the given input image based on color levels using Open CV

PYTHON CODE:

```
import cv2

# Read the image
image = cv2.imread("sample.jpg")

# Check if the image is loaded
if image is None:
    print("Error: Image not found!")
    exit()

# Convert image to grayscale
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Display original image
cv2.imshow("Original Image", image)

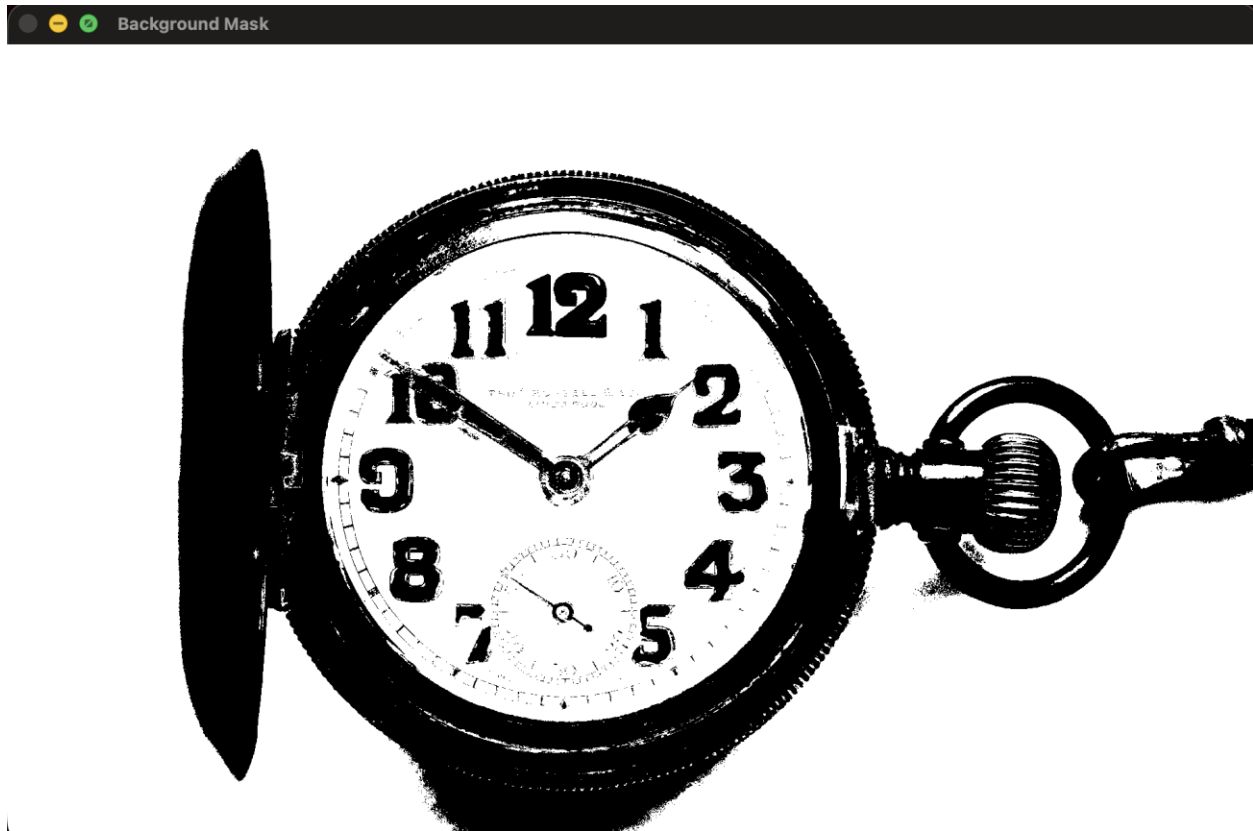
# Display grayscale image
cv2.imshow("Gray-scale Image", gray_image)
```

```
# Wait for a key press
cv2.waitKey(0)

# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:





QUESTION-37:

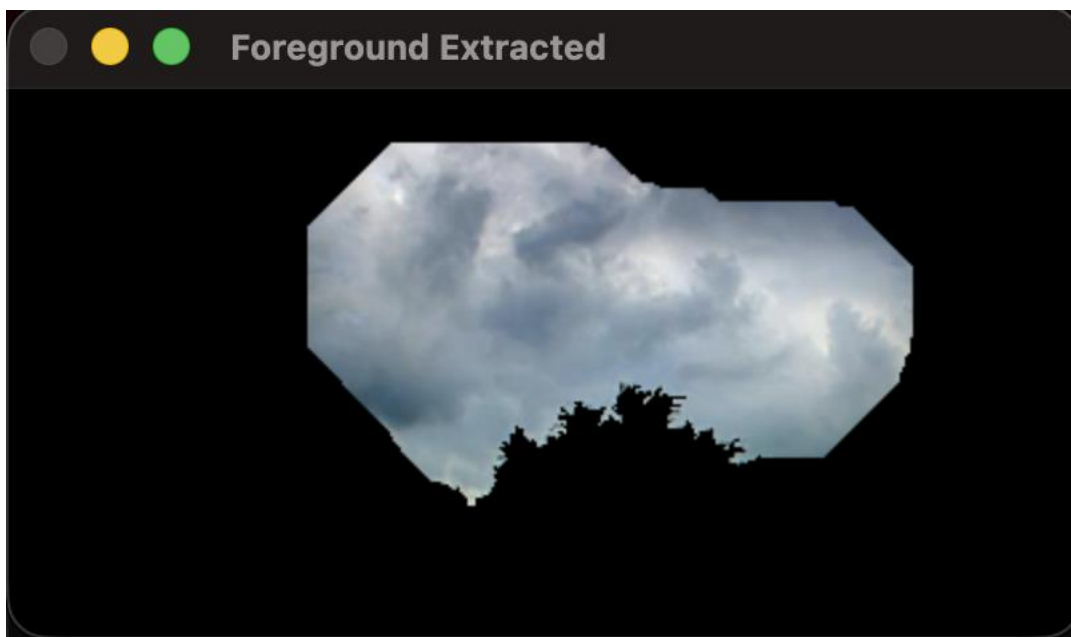
Write a Python function to subtract the foreground of the given input image based on color levels using Open CV.

PYTHON CODE:

```
import cv2
# Read the image
image = cv2.imread("sample.jpg")
# Check if the image is loaded
if image is None:
    print("Error: Image not found!")
    exit()
# Apply Gaussian Blur
```

```
blur_image = cv2.GaussianBlur(image, (15, 15), 0)
# Display original image
cv2.imshow("Original Image", image)
# Display blurred image
cv2.imshow("Gaussian Blur Image", blur_image)
# Wait until a key is pressed
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:





QUESTION-38:

Write a Python function to Count the number of faces for the given input image using Open CV.

PYTHON CODE:

```
import cv2

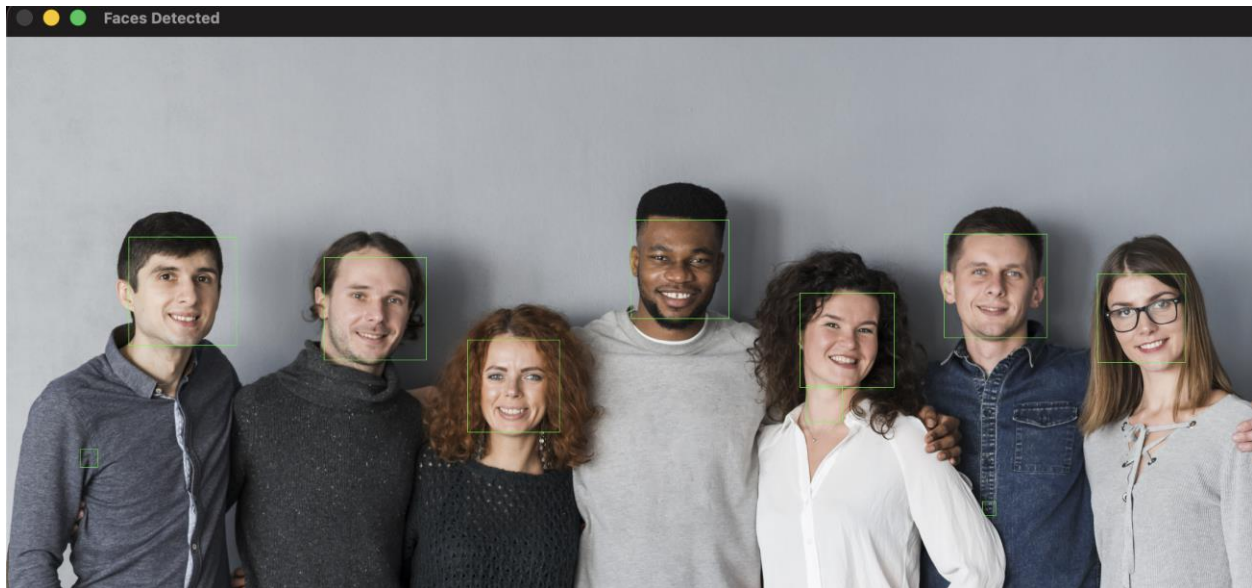
# Read the image
image = cv2.imread("sample.jpg")

# Check if image is loaded
if image is None:
    print("Error: Image not found!")
    exit()

# Convert to grayscale
```

```
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Apply Canny edge detection
edges = cv2.Canny(gray_image, 100, 200)
# Display images
cv2.imshow("Original Image", image)
cv2.imshow("Edge Detected Image", edges)
# Wait for key press
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:



QUESTION-39:

Write a Python function to play the given video in reverse mode in slow

motion.

PYTHON CODE:

```
import cv2  
# Read the image  
image = cv2.imread("sample.jpg")  
# Check if image is loaded  
if image is None:  
    print ("Error: Image not found!")  
    exit ()  
# Convert to grayscale  
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)  
# Apply Histogram Equalization  
equalized_image = cv2.equalizeHist(gray_image)  
# Display images  
cv2.imshow("Original Grayscale Image", gray_image)  
cv2.imshow("Histogram Equalized Image", equalized_image)  
# Wait until a key is pressed  
cv2.waitKey(0)  
# Close all windows  
cv2.destroyAllWindows()
```

OUTPUT:



QUESTION-40:

Write a Python function to extract the text from videos.

PYTHON CODE:

```
import cv2
import matplotlib.pyplot as plt
def analyze_histogram(image_path):
    # Read the image
    image = cv2.imread(image_path)
    # Check if image is loaded
    if image is None:
```

```
    print("Error: Image not found!")
    return
# Display the original image
cv2.imshow("Original Image", image)
# BGR color channels
colors = ('b', 'g', 'r')
# Create histogram plot
plt.figure(figsize=(8, 5))
for i, color in enumerate(colors):
    histogram = cv2.calcHist([image], [i], None, [256], [0, 256])
    plt.plot(histogram, color=color, label=f"{color.upper()} Channel")
plt.title("Color Histogram Analysis")
plt.xlabel("Pixel Intensity")
plt.ylabel("Number of Pixels")
plt.xlim([0, 256])
plt.legend()
# Show histogram
plt.show()
# Wait for key press
cv2.waitKey(0)
cv2.destroyAllWindows()
# Call the function
analyze_histogram("sample.jpg")
```

OUTPUT:

